

Negative Exponent Law*Question Bank 3.04*

Show your work. Circle your final answer.

For questions 1 through 6, rewrite each expression as an equivalent expression with positive exponents. Do not evaluate.

1. 8^{-3}

2. $\frac{1}{(-4)^{-5}}$

3. $(-2)^{-8}$

4. $\frac{1}{10^{-2}}$

5. $-5 \cdot 6^{-4}$

6. $\frac{2^{-3}}{9^{-2}}$

For questions 7 and 8, apply the negative exponent law, then the product or quotient of powers law, and evaluate.

7. Evaluate each expression.

a. $(-3)^2 \cdot (-3)^{-5}$

b. $2^8 \cdot 2^{-4}$

c. $9^9 \cdot 9^{-9}$

8. Evaluate each expression.

a. $(-5)^{-6} \div (-5)^{-4}$

b. $2^0 \div 2^{-3}$

c. $3^{-7} \div 3^5$

Opposite or reciprocal.

9. Four expressions are shown: -4 , 4^{-1} , $(-4)^{-1}$, -4^{-1} .
- a. Give the value of each one.

- b. Which of the four is the reciprocal of 4, and which is the opposite of 4?

Working backwards.

10. Find the value of n that makes each statement true.

a. $2^n = \frac{1}{32}$

b. $10^n = 0.001$

c. $7^3 \cdot 7^n = 7^{-1}$

11. Which is the value of 4^{-3} ?

- ☐ A. $\frac{1}{64}$
- ☐ B. -64
- ☐ C. -12
- ☐ D. $\frac{1}{12}$

12. Select ALL of the expressions whose value is $\frac{1}{64}$.

- ☐ A. 2^{-6}
- ☐ B. 4^{-3}
- ☐ C. 8^{-2}
- ☐ D. $\left(\frac{1}{2}\right)^6$
- ☐ E. -2^6
- ☐ F. 2^{-8}